

Supplementary Material

Corrected Statements and Proofs for the Revised Long-Memory Limit Theorems

Companion to `revised_theorems.tex`; machine-checked items verified in the Lean 4 repository at closure commit 16715db

October 7, 2026

This supplement gives, for each substantive correction identified in the audit of 19-AOS1825.pdf: the original claim, the error (with the counterexample where one exists), the corrected statement, and a proof or proof sketch. Section 3 then proves in full the honest-rate drift proposition underlying the revised main theorem. Full written chains are in `direct_proofs/01–24`; per-item machine status is in `paper_revision/theorem_inventory.md`. Status tags: [Lean] = completely formalized; [written] = written mathematical proof. Proofs marked “English rendering” are faithful translations of the corresponding Chinese sources in `direct_proofs/`; every item also carries a pointer *Full written chain* to the exact source files, which remain authoritative for any detail not reproduced here.

1 Corrections to the main theorems

1.1 Critical-regime variance (Theorem 3.3(ii), via S.3.2)

Original. In the critical regime $H(t) = \frac{3}{4}$ the limiting variance is expressed through the kernel center value. **Error.** The derivation of the diagonal contribution ε_4 in S.3.2 replaces $f^2(i/N)f^2(j/N)$ by $f^4(0)$ for $|i - j|$ small, which is not summable to $f^4(0)$; the correct diagonal sum tends to $\int f^2$, and the Hermite rank-2 factor $2!$ together with the squared denominator of g was mishandled. **Corrected.**

$$\text{Var} \longrightarrow \frac{9}{16} \int \omega_r(x)^2 dx.$$

Proof (English rendering of `direct_proofs/03`, §5–6). For unit-stride first increments of fractional Brownian motion with constant H ,

$$r_H(k) = \frac{1}{2}(|k+1|^{2H} + |k-1|^{2H} - 2|k|^{2H}), \quad r_H(0) = 1.$$

At $H = \frac{3}{4}$ a Taylor expansion gives $r_{3/4}(k) = \frac{3}{8}|k|^{-1/2} + O(|k|^{-5/2})$, so both long-memory spectral coefficients equal $a(\pm 1) = \frac{3}{8}$ and $c_a = 2(\frac{3}{8})^2 = \frac{9}{32}$. The log-square statistic is a weighted quadratic form in the standardized increments; diagonalizing it as $Q_N = \sum_j \lambda_{N,j}(Z_j^2 - 1)$ with $s_N^2 = \sum_j \lambda_{N,j}^2$, the operator bound $\max_j |\lambda_{N,j}| \leq \|B_N\|_{\text{op}} = O(N^{1/2})$ gives $\max_j |\lambda_{N,j}|/s_N = O((\log N)^{-1/2}) \rightarrow 0$, and the independent- χ^2 log-characteristic function

$$\log \mathbb{E} e^{itQ_N/s_N} = \sum_j \left\{ -itu_{N,j} - \frac{1}{2} \log(1 - 2itu_{N,j}) \right\} = -\frac{t^2}{2} + o(1)$$

(with remainder $\leq C_t \max_j |u_{N,j}| \sum_j u_{N,j}^2 \rightarrow 0$) yields the Gaussian limit by Lévy continuity — notably *without* invoking the short-memory Breuer–Major theorem, whose summability hypothesis fails at criticality. The limiting variance is $2c_a \int f^2 = \frac{9}{16} \int \omega^2$.

Counterexample to the printed formula. Take $d = q = p = 1$, $H \equiv \frac{3}{4}$, and the smooth nonnegative bump $K_0(z) = z^2 e^{-1/(1-z^2)}$ on $|z| < 1$ (zero otherwise), normalized to $K = K_0 / \int K_0$, with t interior and $b = n^{-1/2}$. The equivalent kernel $\omega = K / \int K$ then satisfies $\omega(0) = 0$ but $\int \omega^2 > 0$. By fBm stationarity and self-similarity the standardized local increment vectors carry exactly the covariance $r_{3/4}$; the actual weights differ from $N^{-1}\omega(z_i)$ by $N^{-1}e_{N,i}$ with $\sup |e_{N,i}| = O(N^{-1})$, and the covariance bound $|\text{Cov}(g(Y_i), g(Y_j))| \leq Cr(i-j)^2$ makes the weight-perturbation variance $O(N^{-2})$ after critical normalization. Hence the centered $\hat{G}$ converges under the paper’s $\sqrt{N/\log N}$ normalization to a normal law with variance $\frac{9}{16} \int \omega^2 > 0$, while the paper’s kernel-center-value formula returns 0: a counterexample to the *universal critical variance formula itself*, not merely to a step of its derivation.

[written]

Full written chain: `direct_proofs/03`, `direct_proofs/13` (Chinese sources).

1.2 Long-memory sign (Theorem 3.3(iii)–3.4)

Original. The inverted estimator converges to $+Q$. **Error.** The inversion $x \mapsto (c_\sigma - x)/(2 \log n)$ carries a negative slope, so the limit reflects: $-Q$. This is substantive: a legal nonnegative equivalent-kernel instance has third moments of Q and $-Q$ of opposite sign, so the two laws differ. **Corrected.** $2S_n^\psi \log n (\hat{H}_n - h) \Rightarrow -Q$ (Theorem 1 of the companion note). **Proof (direct_proofs/14, §7; the full statement is machine-checked).** The calibration $x \mapsto (c_\sigma - x)/(2 \log n)$ is decreasing, so the expectation-centered limit of $\hat{G}$ transports to the *reflected* limit $-Q$; the affine inversion argument at $a_n = N^\psi$ carries the deterministic center to the true expectation center using $V_n = O(N^{-2\psi})$.

The sign is substantive. Take $p = 2$ and a nonnegative, symmetric, nonzero kernel K ; the local-linear equivalent kernel $\omega = K / \int K$ is then nonnegative, and for any admissible H with $H(t) > \frac{3}{4}$ (constant H suffices) the limiting quadratic-form operator $\mathcal{A}$ is nonnegative and nonzero: using the representation $\frac{1}{2}e^{-a|x-y|} = a \int_{-\infty}^{\min(x,y)} e^{-a(x-u)} e^{-a(y-u)} du$, the ω -weighted quadratic integral is an integral of nonnegative squares, and if it vanished then $F(u) = \int_u^\infty \omega(x) e^{-a(x-u)} dx \equiv 0$ together with $F' = aF - \omega$ would force $\omega = 0$, contradicting $\int \omega = 1$ (all exchanges are justified by the integrability of the kernel family). Hence $\text{tr}(\mathcal{A}^3) > 0$ and

$$\mathbb{E}Q^3 = 8 \text{tr}(\mathcal{A}^3) > 0 \quad \text{while} \quad \mathbb{E}(-Q)^3 < 0,$$

so Q and $-Q$ are not identically distributed — on a submodel allowed by the paper’s assumptions, at the level of the actual statistic’s limit. The fourth cumulant $48 \sum_j \lambda_j^4 > 0$ ($\mathcal{A} \neq 0$) also shows Q is not Gaussian.

[Lean]

Full written chain: `direct_proofs/14`, `direct_proofs/13` (Chinese sources).

1.3 Optimal-bandwidth biased means (Corollary 3.5)

Proof (English rendering of direct_proofs/13, §5–6). Write $\mu_n = \mathbb{E}\hat{G}(t)$, $\theta_n = (c_\sigma - \mu_n)/(2L)$ with $L = \log n$. The file-12 variance bound gives the true variance $V_n = O(a_n^{-2})$ and $\theta_n \rightarrow h$. On the event that the truncation does not act, the *exact* identity

$$2a_n L(\hat{H} - \theta_n) = -a_n(\hat{G} - \mu_n)$$

holds; the truncation event is contained in $\{|\hat{G} - \mu_n| \geq cL\}$, of probability at most $CV_n/L^2 \rightarrow 0$, and $a_n L |\mathbb{E}\hat{H} - \theta_n| \leq Ca_n V_n/L = O((a_n L)^{-1}) \rightarrow 0$, which carries the deterministic center to the true expectation center and keeps the sign.

For the biased means, let p be an integer and let H have a p -th order Peano expansion at t , and set $R_H(t) = \frac{H^{(p)}(t)}{p!} \int z^p \omega(z) dz$. Weight reproduction together with file 12 gives

$$B_n := \mathbb{E}\hat{G} - G_n(h) = -2Lb^p R_H(t) + o(Lb^p) + O(\rho_n(t)).$$

In the short-memory regime $h < \frac{3}{4}$ take $b_0 = (nL^2)^{-1/(2p+1)}$ with $b/b_0 \rightarrow c_0 \in (0, \infty)$, and set $A_n = (n^p/L)^{1/(2p+1)}$; then $A_n L b_0^p = 1$ and $A_n \rho_n(t) \rightarrow 0$ because $\psi > \frac{1}{2} > p/(2p+1)$. Consequently

$$A_n(\hat{G} - G_n(h)) \Rightarrow N(-2c_0^p R_H(t), c_0^{-1} V_h), \quad 2(nL^2)^{p/(2p+1)}(\hat{H} - h) \Rightarrow N(2c_0^p R_H(t), c_0^{-1} V_h),$$

the sign flip coming from the decreasing calibration. When $R = 0$ the additive o -term from the bias expansion is used; no equivalence with 0 is claimed.

[written]; conditional Lean chain connected

Full written chain: `direct_proofs/13`, `direct_proofs/11`, `direct_proofs/01` (Chinese sources).

1.4 Growing-lag remainder (Theorem 8.2(iv))

Counterexample (English rendering of `direct_proofs/04`, §2). Take constant $H = \frac{3}{4}$, $q = 1$, $\sigma = 1$. By stationarity and self-similarity of fBm increments, the paper's standardized difference covariance is exactly $g_1(\frac{3}{4}, u)$, with no n -dependent freezing error. A sixth-order Taylor expansion of $(u \pm 1)^{3/2}$ (odd terms cancelling; coefficients $\binom{3/2}{2} = \frac{3}{8}$, $\binom{3/2}{4} = \frac{3}{128}$; remainder controlled by derivative bounds for $u \geq 2$) gives

$$g_1(\frac{3}{4}, u) |u|^{1/2} = \frac{3}{8} + \frac{3}{128u^2} + O(u^{-4}),$$

so for all $u \geq u_0$ the left side exceeds $\frac{3}{8}$ by at least $\frac{3}{256u^2}$. Now set $b_n = n^{-1/2}$ and $u_n = \lfloor \log n \rfloor$ at a fixed interior t : then $nb_n \rightarrow \infty$, $b_n \log^k n \rightarrow 0$ for every fixed k , $u_n \rightarrow \infty$, $u_n < 2nb_n$, and $t + u_n/n$ stays interior — all the hypotheses of the original 8.2(iv) hold. But

$$\frac{u_n^{-2}}{b_n |\log b_n|} \asymp \frac{\sqrt{n}}{(\log n)^3} \rightarrow \infty \quad (n = e^x \text{ swamps the cubic}),$$

so the remainder cannot be $O(b \log b)$: the original 8.2(iv) fails already on the constant- H submodel allowed by the paper.

Corrected versions. (i) Constant H : $C_n(t, t + u/n) = \{c_q(H) + O(|u|^{-2})\} |u|^{-\psi}$, $\psi = 2q - 2H$. (ii) Nonconstant H (English rendering of `direct_proofs/12`, §3.3): for $2 < |k| \leq 2nb$, from the integral representation, the Hölder increment $H(s + v) - H(s) = O(1/n)$, and the normalization factor $n^{H(u) - H(s)} = 1 + O(|k|L/n)$,

$$C_n(s, s + k/n) = \left\{ c_1(H(s)) + O\left(k^{-2} + \frac{|k|L}{n} + (|k|/n)^{\psi(s)}\right) \right\} |k|^{-\psi(s)}.$$

The mechanism: the power error from the shifted exponent is $O(L/n)$, the derivative distance-logarithm remainder at most $C|k|L/n$, the smooth background contributes $(|k|/n)^{\psi(s)}$, and the frozen integral, written with $V = n(w - v)$ satisfying $\mathbb{E}V = 0$, $|V| \leq 1$, obeys the second-order Taylor bound $\mathbb{E}|k + V|^{-\psi(s)} = |k|^{-\psi(s)}(1 + O(k^{-2}))$. All remainders are absolute coefficient errors, never divided by the possibly-vanishing c_1 .

[written]; interior pieces Lean

Full written chain: `direct_proofs/04`, `direct_proofs/12`, `direct_proofs/18` (Chinese sources).

1.5 Local-polynomial bias (Lemma 8.4)

Original. An integral-region/quadrature argument. **Error.** The quadrature over the integration region is invalid. **Corrected.** Define $M_{kl} = \sum_i K_i A_{ik} A_{il}$ and $w_i = K_i \sum_l (M^{-1})_{0l} A_{il}$. **Proof.** Invertibility of M gives $\sum_i w_i A_{ik} = \delta_{k0}$ exactly; hence for $F_i = \beta_0 + \sum_{k \geq 1} \beta_k A_{ik} + r_i$ with $|r_i| \leq B$ on the support of K ,

$$\left| \sum_i w_i F_i - \beta_0 \right| \leq \sum_i |w_i| B = O(b^p),$$

using the uniform L^1 weight bound. No $(nb)^{-\min(2,p)}$ term is needed.

[Lean](designs, weights, equivalent kernel, bias leading term)

Full written chain: `direct_proofs/02`, `direct_proofs/18` (Chinese sources).

1.6 Vector Z limits (Lemma 8.5)

Corrections. (i) The three limit statements must be made for $Z - \mathbb{E}Z$: the normalization of the estimator chain transports centered vectors, and the bound $\mathbb{E}Z = O((nb)^{d/2}\rho)$ alone does not prove that the normalized $\mathbb{E}Z$ vanishes — if the original uncentered Z is kept, this vanishing must be established separately. (ii) The dimension of the local-polynomial coefficient vector a is the basis dimension $S = \binom{p+d}{d}$ -type count, not the ambient d ; the covariance structure and the limits are stated in $\mathbb{R}^S$. (iii) Kernel conditions ensuring the nondegeneracy used by the limits are added explicitly (they are exactly the ones repaired in S.2.1 below). The three limit statements themselves are at the written level (the finite-Hermite CLT for non-absolutely-summable correlation arrays of file 13 §2 supplies the mechanism); the expectation-bias leading terms derived from them are machine-checked.

expectation-bias leading terms [Lean]; three Z -limits [written]

Full written chain: `direct_proofs/03`, `direct_proofs/13`, `direct_proofs/14` (Chinese sources).

2 Corrections to the supplementary lemmas

2.1 S.1.1 (Fourier adjacent differences)

Diagnosis. As a complex Lebesgue integral the claim fails: near 0 the imaginary part of the integrand behaves like $-ikx/|x|^{2+\delta}$, which is not absolutely integrable.

Corrected statement and proof (English rendering of `direct_proofs/06`, §2). For $0 \leq \delta \leq \frac{1}{2}$ define

$$A_\delta = \int_{\mathbb{R}} \frac{1 - \cos x}{|x|^{2+\delta}} dx, \quad B_\delta = \int_{\mathbb{R}} \frac{(1 - \cos x) \log |x|}{|x|^{2+\delta}} dx.$$

A_δ , and the absolute integral defining B_δ , are uniformly bounded on this δ -range: near 0 they are dominated by $C|x|^{-1/2}$ and $C|x|^{-1/2}|\log |x||$, in the tail by $C|x|^{-2}$ and $C|x|^{-2}\log |x|$. For $k \geq 1$ the substitution $y = kx$ gives the exact scalings

$$I_\delta(k) = \int \frac{1 - \cos(kx)}{|x|^{2+\delta}} dx = A_\delta k^{1+\delta}, \quad J_\delta(k) = \int \frac{(1 - \cos(kx)) \log |x|}{|x|^{2+\delta}} dx = k^{1+\delta} (B_\delta - A_\delta \log k),$$

with $I_\delta(0) = J_\delta(0) = 0$. Assume $\delta_n \geq 0$ and $n^{\delta_n} \rightarrow 1$; then eventually $\delta_n \leq \frac{1}{2}$ and $n^{\delta_n} \leq 2$. For $2 \leq k \leq n$ the mean-value theorem gives $k^{1+\delta_n} - (k-1)^{1+\delta_n} \leq (1+\delta_n)n^{\delta_n} \leq 3$, and the derivative of $x^{1+\delta} \log x$ is $x^\delta[(1+\delta)\log x + 1] = O(\log n)$ on $[1, n]$. With the uniform bounds on A, B ,

$$\sup_{1 \leq k \leq n} |I_{\delta_n}(k) - I_{\delta_n}(k-1)| = O(1), \quad \sup_{1 \leq k \leq n} |J_{\delta_n}(k) - J_{\delta_n}(k-1)| = O(\log n),$$

the case $k = 1$ being handled separately from the boundedness of A_δ , B_δ . This proves the real-part (cosine) repaired version of S.1.1; under symmetric truncation the imaginary part vanishes identically, so the same proof covers the explicitly defined principal-value version.

[written](repair); not formalized

Full written chain: `direct_proofs/06` (Chinese source).

2.2 S.1.2 (filtered Fourier integral)

Status. No counterexample found; a more direct route exists but is not formalized. **Route.** Use the cosine form and the second difference $(k+1)^{1+\delta} - 2k^{1+\delta} + (k-1)^{1+\delta}$ in place of the contour argument; $k = 1$ is handled separately and $\delta = 0$ returns 0; the real part must be retained in the contour derivation.

Full written chain: `direct_proofs/06` (Chinese source).

2.3 S.2.1 (g bounded away from zero)

Error. The hypotheses are insufficient: at $h = 0$ the difference vanishes identically, and without compactness away from 1 there is no uniform positive lower bound — for $q = 2$, $g_2(H, 0, 1) = 4 - 2^{2H} \rightarrow 0$ as $H \uparrow 1$.

Corrected statement and proof (English rendering of `direct_proofs/05`, §2). Fix $q \geq 1$ and $h \neq 0$. For the constant- H model, applying q -th order differences on both sides of the corrected spectral representation (5.2) of §S.2.2 below gives

$$g_q(H, 0, h) = \frac{1}{2D(H)^2} \int_{\mathbb{R}^d} \frac{|1 - e^{i\langle h, \xi \rangle}|^{2q}}{|\xi|^{2H+d}} d\xi > 0 :$$

near 0 the numerator is $O(|\xi|^{2q})$ and the integral converges because $q > H$; the tail converges because $H > 0$; and $h \neq 0$ makes the numerator strictly positive on an open set. If $H \in [\eta, 1 - \eta]$, the same integrable majorant controls the whole family (radially $r^{2q-2(1-\eta)-1}$ near 0 and $r^{-2\eta-1}$ at infinity), and similarly for $D(H)$; dominated convergence makes $H \mapsto g_q(H, 0, h)$ continuous, whence by compactness $\inf_{H \in [\eta, 1-\eta]} g_q(H, 0, h) > 0$. This is a direct repair that does not rely on the conditionally-negative-definite kernel theorem quoted in the original; the two added conditions ($h \neq 0$; compactness away from 0 and 1, available from [A1] rather than [A3]) are both indispensable, as the two counterexamples show.

repaired 1-d version [Lean]

Full written chain: `direct_proofs/05` (Chinese source).

2.4 S.2.2 / S.3.1 (leading singular coefficient)

Proof of the corrected constants (English rendering of `direct_proofs/04`, §1, and of `direct_proofs/05`, §1). (i) *Difference expansion.* Let $q \geq 1$ and $F_H(u) = |u|^{2H}$, and

$$g_q(H, u) = -\frac{1}{2} \sum_{i,j} (-1)^{i+j} \binom{q}{i} \binom{q}{j} F_H(u + i - j).$$

By the binomial expansion (Vandermonde), $g_q(H, u) = \frac{(-1)^{q+1}}{2} \delta^{2q} F_H(u)$ with $\delta F(u) = F(u + \frac{1}{2}) - F(u - \frac{1}{2})$. For $u > 2q$, repeated integration of the fundamental theorem gives $\delta^{2q} F_H(u) = \int_{[-1/2, 1/2]^{2q}} F_H^{(2q)}(u + \sum s_i) ds$; the mean of $S = \sum s_i$ is 0 with $|S| \leq q$, and a first-order Taylor

expansion of $F_H^{(2q)}$ has second-order remainder bounded by $Cu^{2H-2q-2}$ uniformly for H in a fixed compact subinterval. Hence

$$g_q(H, u) = c_q(H)|u|^{2H-2q} + O(|u|^{2H-2q-2}), \quad c_q(H) = \frac{(-1)^{q+1}}{2} \prod_{l=0}^{2q-1} (2H - l),$$

and in particular $c_1(H) = H(2H-1)$: the paper's printed c_h is missing the factor $\frac{1}{2}$. When the product vanishes, (the display) remains an absolute remainder bound — nothing is divided by $c_q(H)$. The far-field form $g = (c_h + o(1))|u|^{-\psi}$ of S.3.1 follows with the same c_h , and its $o(1)$ must not be strengthened to $O(b \log b)$. (ii) *Spectral normalization*. With $D(H)^2 = \int (1 - \cos \xi_1)/|\xi|^{2H+d} d\xi$ (finite and positive for $0 < H < 1$), rotation and scaling give $\int (1 - \cos \langle t, \xi \rangle)/|\xi|^{2H+d} d\xi = D(H)^2 |t|^{2H}$. For $a = \frac{H_t + H_s}{2}$ the absolutely convergent integral $I(t, s) = \int (e^{i\langle t, \xi \rangle} - 1)(e^{-i\langle s, \xi \rangle} - 1)/|\xi|^{H_t + H_s + d} d\xi$ has vanishing imaginary part (odd symmetry) and real part $D(a)^2 \{|t|^{2a} + |s|^{2a} - |t - s|^{2a}\}$, so the correct covariance normalization is $C(t, s) = \frac{\sigma^2}{2D(H_t)D(H_s)} I(t, s)$: omitting the $\frac{1}{2}$ would double the variance to $2\sigma^2 |t|^{2H}$ at $t = s$.

instantiated and machine-verified inside the closure chain [Lean]

Full written chain: `direct_proofs/04`, `direct_proofs/05`, `direct_proofs/12` (Chinese sources).

2.5 S.2.3 (mixed derivatives, near-diagonal)

Counterexample (English rendering of `direct_proofs/12`, §2). Fix an admissible base point (h_0, t_0) with t_0 interior and $h_0 \in (0, 1)$, and take the lawful smooth Hurst function $H_\varepsilon(x) = h_0 + \varepsilon(x - t_0)$ with $\varepsilon > 0$ small (it is smooth and stays away from 0 and 1). Let the smooth background be $P(s, u) = B(H_\varepsilon(s), H_\varepsilon(u))(s^a + u^a)$, where B is the normalizing factor of the representation. B is symmetric with $B(h, h) = \frac{1}{2}$, hence $B_h(h, h) = B_k(h, h) = 0$, and direct differentiation gives

$$\partial_s \partial_u P(t_0, t_0) = \varepsilon t_0^{2h_0-1} (2h_0 \log t_0 + 1) + \varepsilon^2 t_0^{2h_0} \{2B_{hk}(h_0, h_0) + (\log t_0)^2\} = \frac{1}{2} \varepsilon t_0^{2h_0-1} + O(\varepsilon^2) > 0.$$

Fix such an ε . The remainder of the singular part alone, computed in §2 of the file, is $O(r^{1-\psi_0}(1 + |\log r|)) = o(1)$ with $\psi_0 = \frac{1}{2}$, so

$$\lim_{r \downarrow 0} \left\{ \partial_s \partial_u C(t_0, t_0 + r) - c_1(h_0) r^{-\psi_0} \right\} = \partial_s \partial_u P(t_0, t_0) > 0,$$

whereas the original S.2.3 remainder, a *relative* $O(r \log r)$, would force the bracketed difference to tend to 0 — a contradiction. The principal-term expansion as printed therefore fails.

Repair. Add the smooth background of relative size r^ψ : the corrected expansion retains, near the diagonal, the $x^{H(x)+H(y)-q}$ -times-log-power background derivative, and the normalized finite differences on the actual grid are then controlled by it; when $H(s) > \frac{1}{2}$ one has $\psi(s) < 1$ and the term cannot be absorbed into $O(r|\log r|)$. Files 12/13/14 verify that this repair does not change the one-dimensional variance rates or limit types.

[written]; grid normalization pieces [Lean]

Full written chain: `direct_proofs/12`, `direct_proofs/18`, `direct_proofs/04` (Chinese sources).

2.6 S.3.2 (weighted power sums / cycle products)

Corrected jointly with §1.1 above (center value $\int f^2$; $2!$; denominator). The long-memory side — identification of the cycle products with the spectrum of $B = K^{1/2} M_\omega K^{1/2}$ for all orders $k \geq 2$ — is machine-checked.

long-memory identification [Lean]

Full written chain: `direct_proofs/03`, `direct_proofs/13`, `direct_proofs/14` (Chinese sources); the critical side is proved in §1.1 above.

2.7 S.5.1 (spatial average variance of I_1)

Error. The far-distance count misses a factor m : with $H \equiv 7/8$, $q = p = 1$, $b = n^{-1/2}$ one has $\text{Var}(I_1) \approx \log^2 n \cdot n^{-1/2}$ against the claimed $O(\log^2 n \cdot n^{-3/4})$. **Correction.** The count is $m^{2\psi-1}n^{-2\psi}$ times the lag sum; merging weights at $m \approx 1/b$ gives $\text{Var}(I_1) \leq C \log^2 n \cdot T(n)$, which the original bandwidth condition still absorbs into the target term — the main scale rates are preserved.

[written]; the $q = 1$ scale input it feeds is discharged in Lean

2.8 S.5.2 (truncated component I_2')

Proof of the repaired bounds (English rendering of `direct_proofs/16`, §4). The pilot supplies, uniformly in j ,

$$|\mathbb{E}P(s_j) - H(s_j)| \leq C a_n, \quad \mathbb{E}|P(s_j) - H(s_j)|^2 \leq C\{a_n^2 + T_{\bar{\psi}}(N)\}.$$

For $q = 1$ the truncation functional ℓ vanishes identically, so $I_2 = 0$. For $q = 2$, ℓ, ℓ', ℓ'' are bounded on $[0, 1 - \gamma/2]$ and the true H keeps a fixed positive distance from both truncation endpoints. Put $v = \mathbb{E}(P - H)^2$. Projection does not increase the error relative to H , so $\mathbb{E}|\ell(P^\gamma) - \ell(H)|^2 \leq Cv$; moreover $|P^\gamma - P| \leq |P - H|\mathbf{1}_{\{|P-H| \geq d_0\}}$ gives $\mathbb{E}|P^\gamma - P| \leq v/d_0$; and Taylor's formula gives $|\mathbb{E}\{\ell(P^\gamma) - \ell(H)\}| \leq C\{|\mathbb{E}(P - H)| + v\}$. Applying Jensen to the spatial average — with *no* independence assumed across j , hence no fictitious $1/M$ variance gain — yields

$$\mathbb{E}I_2'^2 \leq C\{a_n^2 + T_{\bar{\psi}}(N)\}, \quad |\mathbb{E}I_2| \leq C\{a_n + T_{\bar{\psi}}(N)\},$$

and a_n^2 is absorbed by a_n since $a_n \rightarrow 0$. Combined with the repaired S.5.1 bound and the square-sum inequality this recovers Theorem 4.2's MSE decomposition; the role of the bandwidth condition $L^2T(n) = o(T(nb_1))$ is checked separately in each memory regime (short: $bL^2 \rightarrow 0$; long: $L^2b^{2\bar{\psi}} \rightarrow 0$ by choosing the fixed k of (16.1) large; critical: at most $2bL^2$ if $N \geq \sqrt{n}$, at most $n^{-1/2}L^3/\log N$ otherwise).

[written]+ Full pieces

Full written chain: `direct_proofs/16`, `direct_proofs/08` (Chinese sources).

2.9 S.6.1 (irregular-grid covariance)

Counterexample (English rendering of `direct_proofs/05`, §4). Take one-dimensional standard Brownian motion B , $q = 1$, $H = \frac{1}{2}$; regular grids are admissible special cases of the irregular grids allowed. By the Brownian covariance $\min(s, t)$, $\text{Var}\{B(t + 1/n) - B(t)\} = 1/n$, while $g_1(\frac{1}{2}, 0, 1) = 1$; the difference of the two sides tends to -1 and cannot be $O(\log n/n + 1/n)$. This refutes the unnormalized claim S.6.1(i) as printed. (The scalar asymptotic underlying this counterexample is machine-checked in the repository.)

Correct transformation. For arbitrary grid differences $D_n(t)$ define $W_n(t) = \sigma^{-1}n^{H(t)}D_n(t)$. Bilinearity of the covariance gives the *exact* identity

$$\text{Cov}(D_n(t), D_n(s)) = \sigma^2 n^{-H(t)-H(s)} \text{Cov}(W_n(t), W_n(s)),$$

so if the normalized version states $\text{Cov}(W_n(t), W_n(s)) = g(H(t), u, \varpi(t)) + O(\rho_n)$, the original-difference version must be $\sigma^2 n^{-H(t)-H(s)}\{g(H(t), u, \varpi(t)) + O(\rho_n)\}$ — the error term carries the

same prefactor. On the Brownian regular grid with $u = 0$ the corrected form is an exact equality; for general irregular grids the local covariance approximation still requires separate proof — the identity alone does not provide it.

scalar counterexample [Lean]; full instantiation open

Full written chain: `direct_proofs/05` (Chinese source).

2.10 S.7.2 (dimension scope)

No counterexample; scope remark: the $d = 2, 3$ “minimax optimality” assertion has no corresponding lower bound in the manuscript itself, so it cannot be certified even after repairing the upper-bound route.

[written](route only)

Full written chain: `direct_proofs/02` (Chinese source).

3 Proof of the honest-rate drift proposition

This section proves Proposition 2 of the companion note in full. Work under (R1)–(R6) of the companion note: $\delta_n = n^{-h}$, $S_n = n^{1-h}$, $\psi = 2 - 2h \in (0, \frac{1}{2})$, $c_\sigma = \mathbb{E} \log Z^2$, $\widehat{G}_n$ the unit-weight log-square statistic, $\widetilde{H}_n = (c_\sigma - \widehat{G}_n)/(2 \log n)$, $\widehat{H}_n = T(\widetilde{H}_n)$ clipped to $[0, 1]$.

Proposition 1. $2S_n^\psi \log n |\mathbb{E} \widehat{H}_n - h| \rightarrow 0$, with the three-term bound of the companion note; the second term vanishes exactly under the b -band $(1 - h)^2 < 1 - b$, the other two unconditionally.

Proof. Step 1 (exact expectation identity). For each active row j the coefficient-feature combination satisfies, by the stride identity, the norm decomposition

$$\log \|\text{combo}_j\|^2 = -2H_j \log n + \log \|V_j\|^2,$$

where $H_j = f(s_j)$ is the left-endpoint Hölder value and V_j the normalized varying increment. The Gaussian expectation formula gives

$$\mathbb{E} \widehat{G}_n = \sum_j w_j (-2H_j \log n + \log \|V_j\|^2) + c_\sigma, \quad \sum_j w_j = 1,$$

whence, with c_σ pinned,

$$\mathbb{E} \widetilde{H}_n - h = -\left(\sum_j w_j H_j - h\right) - \frac{1}{2 \log n} \sum_j w_j \log \|V_j\|^2. \quad (1)$$

Step 2 (deterministic side). (i) *Weight contraction.* On the active window $|s_j - t| < \delta_n$, so Hölder regularity gives $|H_j - h| \leq D(1 + M)\delta_n$, and $|\sum_j w_j H_j - h| \leq \sum_j |w_j| |H_j - h| \leq C_{\text{st}} \delta_n^p$ with $p \geq 1$ (the sharper exponent uses the class regularity; $p = 1$ suffices). (ii) *Second-order increment log rate.* Write $V_j = F_k(s_j + \ell)$ -type normalized varying increment with $|k - h| \leq K\ell$, $\ell = 1/n$, $K = D(1 + M)$. The exact kernel identity

$$2\langle \text{frozen, mismatch} \rangle = \Lambda \Phi(m) - \Phi(h), \quad \Phi(q) = (s + \ell)^{2q} - s^{2q} + \ell^{2q}, \quad m = \frac{h+k}{2},$$

with $\Lambda - 1 = -\frac{1}{2} \|F_k(1) - F_h(1)\|^2 = O((k - h)^2)$ (uniform parameter Lipschitz; no smoothness of the normalizing constant needed) and the mixed-power Lipschitz bound $|\Phi(m) - \Phi(h)| \leq C|k - h|\ell$ (valid since $2a > 1$ absorbs the logarithmic sensitivity) yields

$$|\|V_j\|^2 - 1| \leq C(1 + K + K^2 + K^3) \ell^{2-2b} = C n^{-(2-2b)},$$

hence $|\log \|V_j\|^2| \leq 2C n^{-(2-2b)} =: C_{\log} n^{-(2-2b)}$ once small, and $|\sum_j w_j \log \|V_j\|^2| \leq C_w C_{\log} n^{-(2-2b)}$ with $C_w = \sum_j |w_j|$ uniformly bounded.

Step 3 (random side; tail-based transfer). Let $m = \mathbb{E}\tilde{H}_n$. Since $\mathbb{E}\tilde{H}_n \rightarrow h \in (3/4, 1)$ (companion note, center-band discharge), eventually $m \in [d', 1 - d']$ with $d' = (1 - h)/2$. On $\{\tilde{H}_n \notin [0, 1]\}$ we have $|\tilde{H}_n - m| \geq d'$. The unit-weight variance bound and the normalized energy give

$$V_n := \text{Var}(\tilde{H}_n) = \frac{\text{Var}(\hat{G}_n)}{4 \log^2 n} \leq \frac{4 g_{\text{LSV}} U^2 C_2}{4 \log^2 n} S_n^{-2\psi} = O\left(\frac{n^{-4(1-h)^2}}{\log^2 n}\right),$$

and Chebyshev gives $\varepsilon_n := \mathbb{P}(|\tilde{H}_n - m| \geq d') \leq V_n/d'^2$. Since $|T(x) - x| \leq |x - m|$ off $[0, 1]$ and $T(x) = x$ on $[0, 1]$,

$$|\mathbb{E}\hat{H}_n - \mathbb{E}\tilde{H}_n| \leq \mathbb{E}[|\tilde{H}_n - m| \mathbf{1}_{\tilde{H}_n \notin [0, 1]}] \leq V_n + \varepsilon_n \leq V_n(1 + d'^{-2}).$$

Step 4 (exponents). Combining $\mathbb{E}\hat{H}_n - h = [\mathbb{E}\tilde{H}_n - h] + [\mathbb{E}\hat{H}_n - \mathbb{E}\tilde{H}_n]$ with Steps 2–3,

$$2S_n^\psi \log n |\mathbb{E}\hat{H}_n - h| \leq \underbrace{C_1 \frac{\log n}{n^{h-2(1-h)^2}}}_{\text{smoothing}} + \underbrace{C_2 n^{-(2-2b)-2(1-h)^2}}_{\text{mismatch}} + \underbrace{C_3 \frac{1}{n^{2(1-h)^2} \log n}}_{\text{clipping}}.$$

Indeed: the smoothing term is $2S_n^\psi \log n C_{\text{st}} \delta_n^p$ with $S_n^\psi = n^{2(1-h)^2}$ and $\delta_n^p = n^{-hp}$, and $2(1-h)^2 - hp \leq 2(1-h)^2 - h < 0$ since $2(1-h)^2 < \frac{1}{8} < \frac{3}{4} < h$; the mismatch term is $S_n^\psi C_w C_{\log} n^{-(2-2b)}$ with exponent $2(1-h)^2 - (2-2b)$, negative exactly when $(1-h)^2 < 1-b$, i.e. (R4); the clipping term is $2S_n^\psi \log n V_n(1 + d'^{-2}) = O(n^{2(1-h)^2} \log n \cdot n^{-4(1-h)^2} / \log^2 n)$, unconditionally vanishing. $\square$

Remark 2 (Sharpness). The mismatch rate is attained: the expansion of $\log \|V_j\|^2$ has leading coefficient $f'(s)s^{2h-1}[2hD'/D + 2h \log s + 1]$ evaluated at the window, generically nonzero, so no per-row expansion order reaches the rate n^{-h} and the manuscript's original bias envelope at that rate fails for non-constant profiles. The window refutations are machine-checked; the lower-bound statement is derived by hand and registered.

4 Formalization cross-reference

Every [Lean] tag above is a theorem in the repository, axiom-checked within `{propext, Classical.choice, Quot.sound}`, zero sorry; the entry points are `actualQ1_knownScaleH_fullChain_honestRate` (main theorem), `honestRate_drift_tendsto_zero` (Proposition 1), and the audit theorems listed in the companion note's provenance table. Automated re-verification: `verification/final_acceptance.py`.

References

1. Source manuscript under audit: `19-A0S1825.pdf` (provided with the project; not modified). Numbered results 3.1–3.5, 4.1–4.3, 8.1–8.5 and S.1.1–S.7.2 refer to it.
2. Companion notes: `paper_revision/revised_theorems.{tex,pdf}` (the revised main theorem and the validation-status tables) and `paper_revision/theorem_inventory.md` (per-item machine status with Lean theorem names).

3. Full written chains (Chinese, authoritative for details not rendered here): `direct_proofs/01–24`. The per-item mapping is recorded in `results/catalog.json`.
4. J. Bardet, M. Surgailis, *Moment bounds and central limit theorems for Gaussian subordinated arrays*, arXiv:1104.4732 — Lemma 1 is the only external lemma consumed by the Lean chain (as the hypothesis class of `bardetSurgailis_weighted_evenMoment_bound`); its scalar rank-2 special case is used, as declared in `direct_proofs/19`.
5. *The functional Breuer–Major theorem*, arXiv:1808.02378 — cited for contrast only (§1.1): the critical-regime proof here deliberately avoids the Breuer–Major summability hypothesis, which fails at criticality.